4. Describe the electrolysis of molten NaCl using inert electrodes
5. State Faraday's Laws of electrolysis
6. Describe the construction of Daniel cell. Write the cell reaction.
7. Why is anode in galvanic cell considered to be negative and cathode positive electrode?
8. The conductivity of a 0.01M solution of a 1 :1 weak electrolyte at 298K is 1.5×10^{-4} S cm^{-1}.

   i) molar conductivity of the solution

   ii) degree of dissociation and the dissociation constant of the weak electrolyte

   Given that

   $\lambda^{\circ}_{cation} = 248.2$ S cm^2 mol^{-1}

   $\lambda^{\circ}_{anion} = 51.8$ S cm^2 mol^{-1}
9. Which of 0.1M HCl and 0.1 M KCl do you expect to have greater $\overset{0}{\Lambda}_m$ and why?
10. Arrange the following solutions in the decreasing order of specific conductance.

    i) 0.01M KCl ii) 0.005M KCl iii) 0.1M KCl

    iv) 0.25 M KCl v) 0.5 M KCl
11. Why is AC current used instead of DC in measuring the electrolytic conductance?
12. 0.1M NaCl solution is placed in two different cells having cell constant 0.5 and 0.25cm^{-1} respectively. Which of the two will have greater value of specific conductance.
13. A current of 1.608A is passed through 250 mL of 0.5M solution of copper sulphate for 50 minutes. Calculate the strength of Cu^{2+} after electrolysis assuming volume to be constant and the current efficiency is 100%.
14. Can Fe^{3+} oxidises bromide to bromine under standard conditions?

    Given: $E^{\circ}_{Fe^{3+}|Fe^{2+}} = 0.771$

    $E^{\circ}_{Br_2|Br^-} = 1.09$V.
15. Is it possible to store copper sulphate in an iron vessel for a long time?

    Given : $E^{\circ}_{Cu^{2+}|Cu} = 0.34$ V and $E^{\circ}_{Fe^{2+}|Fe} = -0.44$V.
16. Two metals M_1 and M_2 have reduction potential values of -xV and +yV respectively. Which will liberate H_2 and H_2SO_4.
17. Reduction potential of two metals M_1 and M_2 are $E^{\circ}_{M_1^{2+}|M_1} = -2.3$V and $E^{\circ}_{M_2^{2+}|M_2} = 0.2$V Predict which one is better for coating the surface of iron. Given : $E^{\circ}_{Fe^{2+}|Fe} = -0.44$V
18. Calculate the standard emf of the cell: $Cd|Cd^{2+}\|Cu^{2+}|Cu$ and determine the cell reaction. The standard reduction potentials of $Cu^{2+}|Cu$ and $Cd^{2+}|Cd$ are 0.34V and -0.40 volts respectively. Predict the feasibility of the cell reaction.

XII U9 Electro Chemistry.indd 66 16-12-2022 19:37:50